A Technical Introduction to Dropout regularization

$$\tilde{h} = r \odot h, \quad r_i \sim \mathrm{Bernoulli}(p)$$

Section 1

Regularizers for sparsity Assume that a dictionary φ j {\displaystyle \phi _{j}} with dimension p {\displaystyle p} is given such that a function in the function space can be expressed as:

Section 2

where (x ^ i , y ^ i) , 1 ≤ i ≤ n , {\displaystyle ({\hat {x}}_{i},{\hat {y}}_{i}),\,1\leq i\leq n,} would represent samples used for training. In the case of a general function, the norm of the function in its reproducing kernel Hilbert space is:

Section 3

This can be viewed as inducing a regularizer over the L 2 {\displaystyle L_{2}} norm over members of each group followed by an L 1 {\displaystyle L_{1}} norm over groups. This can be solved by the proximal method, where the proximal operator is a block-wise soft-thresholding function:

Section 4

min f ∑ i = 1 n V (f (x ^ i) , y ^ i) + λ ■ f ■ H 2 {\displaystyle \min _{f}\sum _{i=1}^{n}V(f({\hat {x}}_{i}),{\hat {y}}_{i})+\lambda \left\|f\right\|_{\mathcal {H}}^{2}}

Section 5

R (f 1 ■ f T) = ∑ r = 1 C ∑ t ∈ I (r) ■ f t − 1 I (r) ∑ s ∈ I (r) f s ■ H k 2 {\displaystyle R(f_{1}\cdots f_{T})=\sum _{r=1}^{C}\sum _{t\in I(r)}\left\|f_{t}-{\frac {1}{I(r)}}\sum _{s\in I(r)}f_{s}\right\|_{H_{k}}^{2}} where I (r) {\displaystyle I(r)} is a cluster of tasks. This regularizer is similar to the mean-constrained regularizer, but instead enforces similarity between tasks within the same cluster. This can capture more complex prior information. This technique has been used to predict Netflix recommendations. A cluster would correspond to a group of people who share similar preferences.

Section 6

Elastic net regularization tends to have a grouping effect, where correlated input features are assigned equal weights. Elastic net regularization is commonly used in practice and is implemented in many machine learning libraries.

Section 7

In mathematics, statistics, finance, and computer science, particularly in machine learning and inverse problems, regularization is a process that converts the answer to a problem to a simpler one. It is often used in solving ill-posed problems or to prevent overfitting. Although regularization procedures can be divided in many ways, the following delineation is particularly helpful:

Section 8

Regularizers for multitask learning In the case of multitask learning, T {\displaystyle T} problems are considered simultaneously, each related in some way. The goal is to learn T {\displaystyle T} functions, ideally borrowing strength from the relatedness of tasks, that have predictive power. This is equivalent to learning the matrix W : T × D {\displaystyle W:T\times D} .

Section 9

min w ∑ i = 1 n V (x ^ i · w , y ^ i) + λ ■ w ■ 2 2 , {\displaystyle \min _{w}\sum _{i=1}^{n}V({\hat {x}}_{i}\cdot w,{\hat {y}}_{i})+\lambda \left\|w\right\|_{2}^{2},}

Section 10

w T = γ n ∑ i = 0 T − 1 (I − γ n X ^ T X ^) i X ^ T Y ^ {\displaystyle w_{T}={\frac {\gamma }{n}}\sum _{i=0}^{T-1}\left(I-{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {X}}\right)^{i}{\hat {X}}^{\mathsf {T}}{\hat {Y}}}

Section 11

min w ∈ R p 1 n ■ X ^ w − Y ^ ■ 2 + λ ■ w ■ 1 {\displaystyle \min _{w\in \mathbb {R} ^{p}}{\frac {1}{n}}\left\|{\hat {X}}w-{\hat {Y}}\right\|^{2}+\lambda \left\|w\right\|_{1}}

Section 12

S λ (v) f (n) = { v i − λ , if v i > λ 0 , if v i ∈ [− λ , λ] v i + λ , if v i < − λ {\displaystyle S_{\lambda }(v)f(n)={\begin{cases}v_{i}-\lambda ,&{\text{if }}v_{i}>\lambda \\0,&{\text{if }}v_{i}\in [-\lambda ,\lambda]\\v_{i}+\lambda ,&{\text{if }}v_{i}<-\lambda \end{cases}}}

Section 13

Explicit regularization is regularization whenever one explicitly adds a term to the optimization problem. These terms could be priors, penalties, or constraints. Explicit regularization is commonly employed with ill-posed optimization problems. The regularization term, or penalty, imposes a cost on the optimization function to make the optimal solution unique. Implicit regularization is all other forms of regularization. This includes, for example, early stopping, using a robust loss function, and discarding outliers. Implicit regularization is essentially ubiquitous in modern machine learning approaches, including stochastic gradient descent for training deep neural networks, and ensemble methods (such as random forests and gradient boosted trees). In explicit regularization, independent of the problem or model, there is always a data term, that corresponds to a likelihood of the measurement, and a regularization term that corresponds to a prior. By combining both using Bayesian statistics, one can compute a posterior, that includes both information sources and therefore stabilizes the estimation process. By trading off both objectives, one chooses to be more aligned to the data or to enforce regularization (to prevent overfitting). There is a whole research branch dealing with all possible regularizations. In practice, one usually tries a specific regularization and then figures out the probability density that corresponds to that regularization to justify the choice. It can also be physically motivated by common sense or intuition. In machine learning, the data term corresponds to the training data and the regularization is either the choice of the model or modifications to the algorithm. It is always intended to reduce the generalization error, i.e. the error score with the trained model on the evaluation set (testing data) and not the training data. One of the earliest uses of regularization is Tikhonov regularization (ridge regression), related to the method of least squares.

Section 14

When labels are more expensive to gather than input examples, semi-supervised learning can be useful. Regularizers have been designed to guide learning algorithms to learn models that respect the structure of unsupervised training samples. If a symmetric weight matrix W {\displaystyle W} is given, a regularizer can be defined:

Section 15

R (w) = ■ σ (W) ■ 1 {\displaystyle R(w)=\left\|\sigma (W)\right\|_{1}} where σ (W) {\displaystyle \sigma (W)} is the eigenvalues in the singular value decomposition of W {\displaystyle W} .

Section 16

prox R ■ (v) = argmin w ∈ R D ■ { R (w) + 1 2 ■ w − v ■ 2 } , {\displaystyle \operatorname {prox} _{R}(v)=\mathop {\operatorname {argmin} } _{w\in \mathbb {R} ^{D}}\left\{R(w)+{\frac {1}{2}}\left\|w-v\right\|^{2}\right\},}